10. $\frac{1}{1-\frac{4}{3}z^{-1}+\frac{1}{2}z^{-2}}$

(a) $\frac{1}{z^{-1}(1-\frac{1}{2}z^{-1})(1-\frac{1}{3}z^{-1})}$

设原式为 $H(z)$

由式子可得 $H(z)$ 的极点为 $\frac{1}{2}$ 和 $\frac{1}{3}$

∵题目说明 $H(z)$ 稳定

∴ROC为 $|z|>\frac{1}{2}$

∵但 $\lim_{z\to\infty} H(z)=\infty$

∴∞不在ROC中

∴~~H(z)~~ 该系统是非因果系统

(b) $\frac{z-\frac{1}{2}}{z^2+\frac{1}{2}z-\frac{3}{16}}$

设原式为 $H(z)$

$H(z)=\frac{z-\frac{1}{2}}{z^2+\frac{1}{2}z-\frac{3}{16}}=\frac{z-\frac{1}{2}}{(z+\frac{3}{4})(z-\frac{1}{4})}$

由 $H(z)$ 得

$H(z)$ 极点为 $-\frac{3}{4}$ 和 $\frac{1}{4}$

∵题目说明系统稳定

∴ROC为 $|z|>\frac{3}{4}$

∵ $\lim_{z\to\infty} H(z)=0$

∴∞在ROC中

∴该系统是因果系统

cc) $\frac{z+1}{z+\frac{4}{3}-\frac{1}{2}z^{-2}-\frac{2}{3}z^{-3}}$

设原式为 $H(z)$

$$H(z)=\frac{6\cdot(z+1)\cdot z^3}{(3z+4)(2z^2-1)}$$

$H(z)$ 极点为 $-\frac{4}{3}$ 和 $\pm\frac{\sqrt{2}}{2}$

∵ 系统稳定

∴ ROC为 $\frac{\sqrt{2}}{2}<|z|<\frac{4}{3}$

∴ 该系统为非因果系统。

10.21

(a) $x[n] = \delta[n+5]$

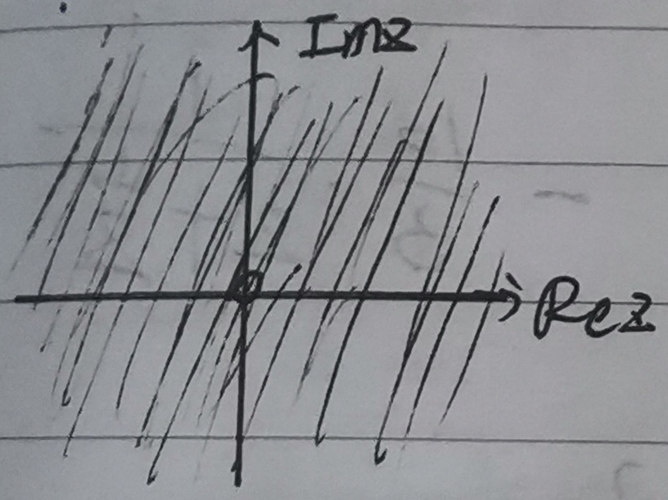
$X(z) = \sum_{n=-\infty}^{\infty} \delta[n+5] \cdot z^{-n}$

$= z^5$ (零点为0，无极点)

∴ ROC为 $0 \le |z| < \infty$

∴ 傅里叶变换存在

零极点图：



(整个z平面，除∞)

(g) $x[n] = 2^n u[-n] + (\frac{1}{4})^n u[n-1]$

$X(z) = \sum_{n=-\infty}^{0} 2^n \cdot z^{-n} + \sum_{n=1}^{\infty} (\frac{1}{4})^n \cdot z^{-n}$

$= \sum_{n=0}^{\infty} 2^{-n} \cdot z^n + \sum_{n=1}^{\infty} (\frac{1}{4})^n \cdot z^{-n}$

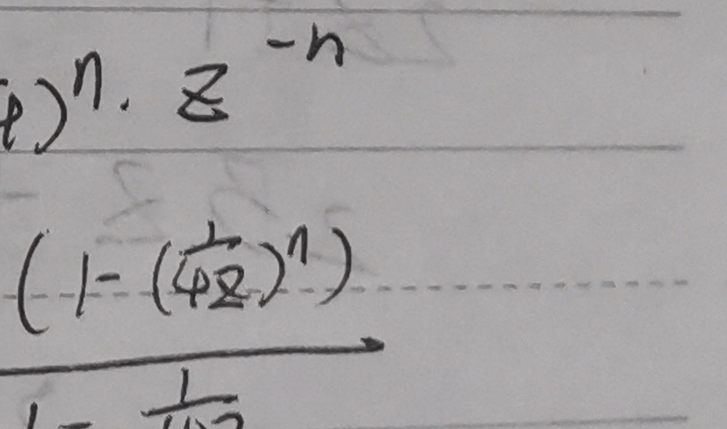
$= \frac{1 \cdot (1-(\frac{z}{2})^n)}{1-\frac{z}{2}} + \frac{\frac{1}{4z} \cdot (1-(\frac{1}{4z})^n)}{1-\frac{1}{4z}}$

∴ ROC为 $\frac{1}{4} < |z| < 2$ ∴傅里叶变换存在

$X(z) = \frac{1}{1-\frac{z}{2}} + \frac{\frac{1}{4z}}{1-\frac{1}{4z}} = \frac{2}{2-z} + \frac{1}{4z-1} = \frac{8z-2+2-z}{(4z-1)(2-z)}$

$= \frac{7z}{(4z-1)(2-z)}$

零极点图：



零点：$z=0$

极点：$z=\frac{1}{4}$，$z=2$

10.24

(a) 部分分式展开法

$X(z)=\frac{1-2z^{-1}}{1+\frac{3}{2}z^{-1}+z^{-2}}$，$x[n]$是绝对可和的

解：$X(z)=\frac{2z^2-4z}{(2z^2+5z+2)})=\frac{2z^2-4z}{(2z+1)(z+2)}$

极点为 $z=-\frac{1}{2}$ 和 $z=-2$

∴$x[n]$是绝对可和的

∴ROC为 $|z|>2$

$$X(z)=\frac{-\frac{10}{3}z}{2z+1}+\frac{\frac{8}{3}z}{z+2}$$

$$=\frac{5}{3}\cdot\frac{1}{1+\frac{1}{2z}}+\frac{8}{3}\cdot\frac{1}{1+\frac{2}{z}}$$

$$=\frac{-10}{3\times2}\cdot\frac{1}{1+\frac{1}{2z}}+\frac{8}{3}\cdot\frac{1}{1+\frac{2}{z}}$$

$$\therefore x[n]=(-\frac{1}{2})^n u[n]\cdot-\frac{5}{3}$$

$$+\frac{8}{3}\cdot(-2)^n u[n]$$

(b) 长除法

$X(z) = \frac{1-\frac{1}{2}z^{-1}}{1+\frac{1}{2}z^{-1}}$ x[n]为右边序列.

解:

$$1+\frac{1}{2}z^{-1}\overline{)\,1-\frac{1}{2}z^{-1}}\quad \Big|\quad 1-z^{-1}+\frac{1}{2}z^{-2}-\frac{1}{4}z^{-3}\cdots$$

$$1+\frac{1}{2}z^{-1}$$

$$-z^{-1}$$

$$-z^{-1}-\frac{1}{2}z^{-2}$$

$$\frac{1}{2}z^{-2}$$

$$\frac{1}{2}z^{-2}+\frac{1}{4}z^{-3}$$

$$-\frac{1}{4}z^{-3}$$

$$-\frac{1}{4}z^{-3}-\frac{1}{8}z^{-4}$$

$$\frac{1}{8}z^{-4}$$

图

ROC为 $|z|>\frac{1}{2}$

$\therefore X(z) = 1 - z^{-1} + \frac{1}{2}z^{-2} - \frac{1}{4}z^{-3}\cdots$

$= \delta[n] + 2\left(-\frac{1}{2}\right)^n u[n-1]$

10.33

(a) $y[n]-\frac{1}{2}y[n-1]+\frac{1}{8}y[n-2]=x[n]$

求两边Z变换得

$Y(z)(1-\frac{1}{2}z^{-1}+\frac{1}{8}z^{-2})=X(z)$

$H(z)=\frac{Y(z)}{X(z)}=\frac{1}{1-\frac{1}{2}z^{-1}+\frac{1}{8}z^{-2}}$

$=\frac{8z^2}{8z^2-4z+1}$

$=\frac{8z^2}{(z-(\frac{1}{4}+j\frac{\sqrt{15}}{8}))(z-(\frac{1}{4}-j\frac{\sqrt{15}}{8}))}$

∴ 系统为因果

∴ ROC为 $|z|>\frac{1}{2}$

(b) $x[n]=(\frac{1}{2})^n u[n]$

$X(z)=\frac{1}{1-\frac{1}{2}z^{-1}}$ ROC: $|z|>$

$Y(z)=H(z)\cdot X(z)$

$=\frac{8z^2}{(8z^2-4z+1)(1-\frac{1}{2}z^{-1})}=\frac{8z^2\cdot 2z}{(8z^2-4z+1)(2z-1)}$

$=\frac{2z^3}{(8z^2-4z+1)(2z-1)}=\frac{8z}{8z^2-4z+1}+\frac{8z}{2z-1}$

$=\frac{2\cdot\frac{8}{\sqrt{15}j}z}{z-(\frac{1}{4}+j\frac{\sqrt{15}}{8})}+\frac{-\frac{8}{\sqrt{15}j}z}{z-(\frac{1}{4}-j\frac{\sqrt{15}}{8})}+\frac{1}{1-\frac{1}{2}z}$

∴ $y[n]=\frac{8}{\sqrt{15}j}\cdot(\frac{1}{4}+j\frac{\sqrt{15}}{8})^n u[n]-\frac{8}{\sqrt{15}j}(\frac{1}{4}-j\frac{\sqrt{15}}{8})^n u[n]+(\frac{1}{2})^n u[n]$

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10.3]

(a) 由图得

~~$\begin{cases} P(z) = \\ Y(z) = P(z) \end{cases}$~~

$$\begin{cases} P(z) = X[n] + \frac{2}{9} z^{-2} P(z) - \frac{1}{3} z^{-1} P(z) \\ Y(z) = P(z) - \frac{9}{8} z^{-1} P(z) \end{cases}$$

得 $\frac{Y(z)}{X(z)} = \frac{1 - \frac{9}{8} z^{-1}}{1 - \frac{2}{9} z^{-2} + \frac{1}{3} z^{-1}}$ ①

$\Rightarrow (1 - \frac{2}{9} z^{-2} + \frac{1}{3} z^{-1}) Y(z) = (1 - \frac{9}{8} z^{-1}) X(z)$

$\Rightarrow y[n] + \frac{1}{3} y[n-1] - \frac{2}{9} y[n-2] = x[n] - \frac{9}{8} x[n-1]$

(b) $H(z) = \frac{Y(z)}{X(z)} = \frac{1 - \frac{9}{8} z^{-1}}{1 - \frac{2}{9} z^{-2} + \frac{1}{3} z^{-1}}$

$= \frac{72z^2 - 81z}{72z^2 - 16 + 24z}$

$= \frac{72z^2 - 81z}{72z^2 - 16 + 24z}$

① $= \frac{72z^2 - 81z}{72(z - \frac{1}{3})(z + \frac{2}{3})}$

极点 $z = \frac{1}{3}$、$z = -\frac{2}{3}$

因为因果性 $\therefore |z| > \frac{2}{3}$

$\therefore$ 该系统稳定

10、42 ~~(b)~~

(b) $y[n]\theta - \frac{1}{2}y[n-1] = x[n] - \frac{1}{2}x[n-1]$

$x[n] = u[n]$

$y[-1] = 0$

取单边Z变换

~~$Y(z) - \frac{1}{2}(Y(z) + y[-1])z^{-1} = 1 + \frac{1}{2}z^{-1}$~~

$Y(z) - \frac{1}{2}y[-1] - \frac{1}{2}z^{-1}Y(z)$

$= X(z) - \frac{1}{2}x[-1] - \frac{1}{2}z^{-1}X(z)$

代入 $x[-1] = 0$

得 $Y(z) - \frac{1}{2}y[-1] - \frac{1}{2}z^{-1}Y(z)$

$= X(z) - \frac{1}{2}z^{-1}X(z)$

$\therefore Y(z) = \frac{\frac{1}{2}y[n-1]}{1-\frac{1}{2}z^{-1}} + X(z)$

$\because y[n-1] = 0$

$\therefore$ 零输入响应 $= 0$

零状态响应 $= x[n] = u[n]$